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Comprehensive Feature-Driven PCOS Predictor: A Reinforcement Learning-Based Binary Equilibrium Optimization Approach

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Abstract

Poly-cystic ovary syndrome (PCOS) is a prevalent condition that impinges women in their prime reproductive years. Its primary trait is the elevated levels of androgen, a male hormone in female body and leads to several symptoms, including ovarian cysts, irregular menstruation cycles, infertility, obesity, and excessive hair development. Since PCOS exact cause is unknown and its symptoms are uncertain, a timely and accurate diagnosis is essential. In these situations, a (ML) Machine Learning-based PCOS prediction model aids in the diagnostic procedure, addresses time constraints and potential inaccuracies. A plethora of feature selection techniques have been built to discover the best optimal features for the classification of medical dataset. However, finding an efficient solution is still difficult due to noise and redundant information which may degrade the model performance. In this article, a hybrid filter-wrapper approach is proposed to identify the optimal attributes. An ensemble filter method is built, the union of top k attributes from individual filters is considered for further process. Then, Reinforcement Learning-based Binary Equilibrium Optimizer is used to find the reduced optimal features. Here, RL uses SARSA (State–Action–Reward–State–Action) to increase the population diversity in the search space, to avoid the problem of local optima, finally balances the exploration capability during the search. Then, identified optimal features are given as input to BMFK (Bonferroni Mean Fuzzy KNN) classifier and achieved 96.32% of accuracy. Further, several machine learning classifiers have been employed for the prompt diagnosis of PCOS. Finally, Random Forest has produced the highest accuracy of 95.62% than other models.

Keywords Poly-cystic ovary syndrome · Machine learning · Reinforcement learning · Feature selection · SARSA

1 Introduction

The most common endocrine disorder PCOS is characterized by reproductive hormone abnormalities in the female body. PCOS treatment duration is determined by the patient's symptoms. The syndrome identified by the cyst develops on the ovaries of some affected individuals, yet these are neither a common symptom nor the main cause of the illness. Women of reproductive age who were classified as assigned female at birth (AFAB) are susceptible to PCOS, a common hormonal condition [1]. Ovulation cyst, acne, hirsutism, irregular menstruation, and elevated testosterone levels are among the possible side effects. Although the precise etiology of PCOS is yet not known, inflammatory processes, genetics and insulin resistance may be contributing factors. PCOS



also can raise the risk of diabetes, hypertension, depression and other health issues. In the vicinity of 5–10% of women of the age 15 to 49 have PCOS [2]. Almost 5 million women worldwide are affected and 70% of them are often misdiagnosed. Diagnose rates vary by ethnicity, with white women accounting for 4.8% of cases, African American women for 8%, Spanish women for 6.8%, and Asian women for 31.3% have PCOS [3, 4] which has a major influence on their lifestyle and may be the cause of conditions including obesity, anxiety and sleep apnea. This raises the possibility of getting ailments like type 2 diabetes, endometrial cancer and heart problems. This significance makes early diagnosis essential.

A comprehensive assessment of medical history, physical examinations, blood tests and ultrasound imaging is usually core of the PCOS diagnostic process. Eliminating other possible illnesses that can present with similar symptoms is also crucial. But these diagnostic processes can take a long time. Patients and medical professionals are prone to investing significant effort. Moreover, the dependence on manual evaluations presents the likelihood for human mistake, potentially resulting in imprecise diagnosis. It is crucial to ensure accuracy when identifying PCOS and more broadly, when diagnosing any medical problem. Accurate and prompt diagnosis is the cornerstone of efficient medical interventions, allowing medical practitioners to create individualized treatment regimens. In such cases, machine learning model can assist physicians in the accurate and early detection of disease, also help in timely treatment process. It can easily identify the pattern by analyzing and detecting the correlation among the features. Feature Selection (FS) methods play a vital role in identifying pertinent features, and it improves the model prediction accuracy by removing the irrelevant data and reducing the number features [5]. The individual filter methods like chi-square, information gain, mRmR, and Relief use ranking approach to remove the irrelevant features. Also, a metaheuristic approach i.e., subset selection method eliminates redundant features and obtains a small feature set using various optimization algorithms which include Whale Optimization Algorithm (WOA), EO (Equilibrium Optimizer), (PSO) Particle Swarm Optimization, (GWO) Grey Wolf Optimizer, Genetic Algorithm (GA), Ant Colony Optimization (ACO), and Al-Biruni Earth Radius (BER) [6, 7]. Though these algorithms effectively select the optimal reduced feature set by removing irrelevant and redundant features, it is computationally expensive. As each individual filter and wrapper method has its own advantages and limitations, no single method can find the optimal features set from a larger feature set. No-Free-Lunch theorem [8] states that not all FS issues can be resolved by a single optimization strategy. In such scenarios, combining metaheuristic methods helps extract significant features. The metaheuristic algorithms like Grey Wolf Optimization (GWO) and Particle Swarm Optimization (PSO) have been hybridized with the Binary Equilibrium Optimizer (BEO). However, these hybrids faced two major issues, such as static balance (fixed exploration–exploitation parameters cannot adapt to search progress) and premature convergence (population diversity collapses early, resulting in suboptimal feature subsets).

The aforementioned limitations inspired us to introduce a hybrid filter-wrapper method which combines ensemble filter and Reinforcement Learning-based enhanced Binary Equilibrium Optimizer as a wrapper method to obtain optimal feature set. Inspired by mass balance, a physics-based metaheuristic method Equilibrium Optimizer (EO) has been chosen. Several researchers employed the EO algorithm to address a range of issues, such as FS problems [5] and image segmentation [9]. Bonferroni Mean Fuzzy KNN (BMFK) classifier [5] has been used as an evaluation model in wrapper method due to its weighted voting based on mean values instead of nearest neighbor.

The key contributions of this article are as follows:

- A novel hybrid filter-wrapper approach EF-RLBEO (Ensemble Filter—Reinforcement Learning-based enhanced Binary Equilibrium Optimizer) is proposed to find the optimal feature subset.
- Ensemble Filter method is developed to remove irrelevant attributes and find the top ranked reduced feature subset.
- A wrapper method: Reinforcement Learning-based enhanced Binary Equilibrium Optimizer (RLBEO) to improve the convergence speed and address the local optima is implemented using BMFK classifier.
- Random Forest, Decision Tree, SVM, Naïve Bayes, KNN, and AdaBoost have been employed for the early detection of PCOS.

- Accuracy, precision, recall, F1 score, and AUC score have been used to analyze the classifier performance.

Rest of the article is organized as follows: Sect. 2 presents the previous works on hybrid filter wrapper and metaheuristic algorithms. Section 3 illustrates the methods used in this work and Sect. 4 represents the proposed work. Experimental settings and results are discussed in Sect. 5. Conclusion of the research work is given in Sect. 6.

2 Related Works

Effective FS (Feature Selection) techniques are crucial to achieve a high-quality feature subset, which eventually results in improved predictive accuracy and decreased computational complexity. Recent developments in FS techniques are covered in this section with an emphasis on both wrapper and hybrid methods. To handle feature selection problems efficiently, many metaheuristic algorithms have been developed over the past few decades. Wrapper-based FS methods are the main applications for these algorithms, which have gained prominence due to their efficiency.

An open-source dataset with numerical parameters linked to PCOS was used in the research work. Three techniques were used to optimize feature selection: Threshold-driven Optimized Mutual Information Method (TOMIM), Optimized Salp Swarm (OSSM), and Threshold-driven Optimized Principal Component Analysis (TOPCA). To determine the most important characteristics for PCOS prediction, these techniques were refined. The most successful of these was TOMIM, which identified 17 crucial characteristics that allowed the RF (Random Forest) model to attain an astounding accuracy of 99.31% across 25 runs [10]. According to the investigation, the novel feature selection technique, a hybrid PODBoost method improved by the Grey Wolf Optimization (GWO)-based feature selection achieves high accuracy of 97.42% than other methods [11]. Greylag Goose Optimization (GGO) was used to refine the feature selection for the improved classification accuracy of lung cancer. Initially, data preprocessing was carried out using the operations, such as scaling, handling gap factor, and normalization. For the classification, MLP model was employed along with feature set obtained by GGO and achieved the highest accuracy of 98.4%. Finally, results of feature selection and classification were assessed using Wilcoxon signed-rank test and ANOVA test [12]. A Snake optimization algorithm has been used to identify the important characteristics from high-dimensional datasets (cardiovascular data). Five different machine learning classifiers have been utilized for the classification, among all, SO-RF has achieved the highest accuracy of 99.9% of accuracy [13]. BBFS (Binary Breadth-First Search) has been used to identify the pertinent features and achieves an average error rate of 47.29%, which is comparatively lesser than other optimization algorithms, such as Binary Grey Wolf Optimizer (BGWO), Binary Particle Swarm Optimization (BPSO), and Binary Whale Optimization Algorithm (BWAO). Six different machine learning classifiers including NBC, RF, DC, ET, SGD, and QDA were employed, and RF has attained the highest accuracy of 91.4% than other models [14]. The authors used SMOTE to balance the dataset and Pearson correlation for feature selection. For the classification, LSTM and CNN models were used. The better accuracy of 96.59% has been obtained by CNN model than LSTM model [15]. Chi-square SelectKBest was used for feature selection process and selected 8 top ranked features. Random Forest, Decision Tree, Gradient Boosting, KNN, Logistic Regression, and SVM were employed as classifier models. Further to improve the accuracy, hybrid classifier model has been implemented and achieved the highest accuracy of 98.7% [16]. BFE (Backward feature Elimination) has been used for feature selection and selected 30 features. The authors have proposed hybrid model for the classification which combines SMOTE with stacking model and achieved 90.24% of highest accuracy [17]. Pearson correlation with different threshold values for selecting the top ranked features was used and various machine learning models including SVM, Random Forest, Logistic Regression, AdaBoost, XGBoost, and LDA. The highest accuracy of 92.4% was achieved by Random Forest with the threshold of 0.8 [18]. The filter, wrapper, and embedded feature selection for extracting the important features has been employed [19]. The top 28 features produced by embedded method were used for model training. Among various machine learning algorithms, Ensemble RF has outperformed other models with 98.89% of accuracy. To

address the outliers and class imbalance, SMOTE and ENN have been used. 23 top features were identified using chi-square and ANOVA techniques [20]. The experimental results have shown that Extreme Gradient Boosting algorithm has produced 96.03% of accuracy than other models. To identify the important features, GA (Genetic Algorithm) was used. The authors have employed PYCARET and WEKA tool to compare the performance of various models and observed that Extra Tree classifier produced 88% of the highest accuracy with the identified attributes [21]. The enhanced binary Equilibrium method was applied to extract the pertinent features to enhance the model performance. To attain the best exploration process and improve the exploitation stage, opposition-based learning and local search algorithm have been used. SVM (Support Vector Machine) and KNN (K-Nearest Neighbor) were used as classifiers. As compared with PSO, WOA, GOA, DA, GWO, and enhanced SSA methods, the features selected by enhanced Equilibrium Optimizer have obtained the highest accuracy [22]. General Learning Strategy (GLEO) was collaborated with Equilibrium Optimizer to address the local optima issue and improve the capability of identifying important regions. The proposed model has been evaluated using 16 different datasets and observed that it achieves the highest accuracy than SCA RF, BOA, PSO, and GWO [23]. For the feature selection, HHO (Horse Herd Optimization) algorithm was used. The transfer functions S-, U-, and V-shaped were applied to make the solution into binary form. Further to improve the performance of the algorithm, three crossover operators were combined. Compared with all, HOA with S-Shaped and one point crossover surpassed other models [24].

A hybrid EO with bi-phase mutation to obtain the pertinent features has been proposed. In this method, sine factor has been used with Equilibrium to enhance the exploitation. The authors employed Random Forest, SVM, Discriminant Analysis, and KNN as a classifier model. They observed that RF produced the highest accuracy of 97.02% on MED-NODE image dataset [25]. A hybrid FS method enhanced EO algorithm with levy-flight and Entropy-based operator has been introduced. The features produced by mRmR filter method and enhanced EO were used as an input to the wrapper method. They have evaluated the performance of the proposed method on 18 benchmark datasets [26]. The wrapper feature selection method with enhanced EO algorithm was implemented to identify important features. They used chaos theory to improve the convergence speed and address the local optima issue. Fifteen benchmark and four large NLP datasets were used to demonstrate the efficiency of proposed algorithm. They proved that the highest classification accuracy can be achieved with minimal optimal feature subsets [27]. The multi-objective hybrid filter-wrapper method with (WOA) Whale Optimization Algorithm was proposed [28]. In this work, 12 benchmark datasets have been used to demonstrate the efficacy of proposed algorithm. The results show that compared with seven other algorithms, with smaller features, the proposed work obtained better results. The hybrid approach which combines the result of two filter methods and one wrapper method has been introduced [29]. They used filter method to examine the performance of the whole population and employed local search method to find the best features. The results shown that compared with existing works, the proposed work achieves the highest accuracy. Ensembles filter method which combines symmetric uncertainty, chi-square, and ReliefF has been developed. To reduce the computational time of wrapper method, the authors have employed Genetic Algorithm (GA) [30]. The features selected by hybrid method produced the highest accuracy than other methods. The classifiers, such as multilayer perceptron, SVM and KNN, were used.

It was found from the literature that though metaheuristic methods are efficient in identifying pertinent features, improving the convergence speed and addressing the local optima issue is still a challenging task. In this article, Reinforcement Learning-based on-policy algorithm SASRA is integrated with BEO (Binary Equilibrium Optimizer) to balance the exploration and exploitation capability. Additionally, hybridizing filter-wrapper approach is performed to obtain the reduced optimal feature set.

3 Methods

3.1 EO (Equilibrium Optimizer)

To express a mass balance equation of EO method [31], a first-order differential equation has been used as described in Eq. (1)

$$M_{dt}^{dC} = VC_{es} - VC + G \quad (1)$$

where M represents the system concentration, M_{dt}^{dC} indicates the system mass change with volume M and time t . C_{es} denotes the equilibrium state's system concentration, V denotes the system's volumetric flow rate. G defines the system generation rate. When, $M_{dt}^{dC} = 0$, it states that a stable state is attained. The concentration C is determined by updating Eq. (1) as follows Eq. (2).

$$C = C_{es} + (C_0 - C_{es}) \cdot \bar{X} + \frac{G}{\alpha M} (1 - \bar{X}) \quad (2)$$

Here, the initial concentration is denoted as C_0 and X is calculated using Eq. (3),

$$X = \exp[-b(t - t_0)] \quad (3)$$

where t_0 indicates the initial time.

3.1.1 Equilibrium Pool and Particles Initialization

Equation (4) with random initialization is used to create the initial particle concentration in the designated space.

$$C_i = C_{\min} + \text{random}_i(C_{\max} - C_{\min}) \dots \dots \dots i = 1, 2, 3, 4, \dots \dots, n \quad (4)$$

Here, the min and max values of dimensions are represented as $C_{\min}$ and $C_{\max}$ and random vector in the range of $[0, 1]$ is represented as random_i , the initial concentration as C_i .

In this approach, equilibrium state is created with five particles along with its vector. These particles are used to balance the ability of exploitation and exploration of the algorithm. To improve the exploration, it finds the first four best particles and the arithmetic mean of these four particles is a fifth particle, which enhances the capability of exploitation Eqs. (5) and (6). To create an equilibrium pool vector, these five particles are used.

$$C_{es, \text{pool}} = \{C_{es1}, C_{es2}, C_{es3}, C_{es4}, C_{es(\text{avg})}\} \quad (5)$$

$$C_{es(\text{avg})} = \frac{(C_{es1}, C_{es2}, C_{es3}, C_{es4}, C_{es(\text{avg})})}{4} \quad (6)$$

3.1.2 Generation Rate (G) and Exponential Term (X)

In EO, the exponential term (X) has been used Eqs. (7) and (8) to achieve a delicate balance between the particles capacity for exploration and exploitation.

$$X = d_1 \text{sign}(\vec{r}\vec{v} - 0.5) \left[e^{-\vec{\alpha}t} - 1 \right] \quad (7)$$

$$t = \left(1 - \frac{\text{Irv}}{\text{Irv}_{\max}}\right)^{\left(d_2 \times \frac{\text{Irv}}{\text{Irv}_{\max}}\right)} \quad (8)$$

where the maximum number of iterations and current iteration value is represented as $\text{Irv}_{\max}$ and Irv . d_1 and d_2 defines the constant value of 2 and 1. The random vector between $[0, 1]$ denoted as rv .

To obtain the optimal solution with enhanced exploitation capability, Generation rate (G) has been used with first-order exponential term in EO algorithm. Equation (9) represents the generation rate.

$$\overline{G} = \overline{G}_0 e^{-\alpha(t-t_0)} = \overline{G}_0 \overline{X} \quad (9)$$

where the initial generation rate is denoted as G_0 Eq. (10) and decay constant is represented as α

$$\overline{G}_0 = \overline{t}_p (C_{\text{es}} - \alpha C) \quad (10)$$

$$\overline{t}_p = \begin{cases} 0.5v_1, & v_2 \geq G_p \\ 0, & v_2 < G_p \end{cases} \quad (11)$$

Here $G_p = 0.5$ is used to achieve the balanced exploration and exploitation. $\overline{t}_p$ indicates the parameter to constrain the Generate rate Eq. (11), v_1 and v_2 represents the random values between $[0, 1]$.

Finally, to update the particles concentration using EO is given in Eq. [12].

$$C(\text{Irv} + 1) = C_{\text{es}} + (C - C_{\text{es}}) \cdot \overline{X} + \frac{G}{\alpha M} (1 - \overline{X}) \quad (12)$$

Here, to obtain more accurate solution, Equilibrium concentration, enhancing exploration and exploitation ability has been combined. M is set to 1.

3.2 Ensemble Filter Methods

An EF (Ensemble Filter) is a robust and successful feature selection approach, which identifies reliable and efficient feature subsets by combining the advantage of several filter methods. The individual filter method computes the relevance of each feature based on their statistical relationship with the target value. To identify pertinent features, these methods evaluate the importance by assigning a rank to each feature based on correlations. In this article, Chi-square, Pearson, Relief, and Mutual Information have been employed as individual filter methods. The top k features from each method are combined to obtain optimal features.

3.3 Reinforcement Learning Algorithm—SARSA

Reinforcement Learning is a machine learning paradigm in order to obtain maximum reward points. It trains the agents to take successive decisions in a given environment. Here, the main goal of the agent is to discover the successful action with an environment via trial and error. In feature selection process, states represent the features assigned with two possible actions of either selecting the feature or not. Finally, the agents receive reward points from the environment based on the classifier accuracy.

To find the best behavior, RL agent iteratively updates its assessment of the state–action pairings (Q value function). In this article, on-policy algorithm SARSA is employed to improve the exploration and exploitation capability. It enhances the Binary Equilibrium Optimizer (BEO) method by modeling the behavior of each candidate as a reinforcement learning agent. In SARSA, the update of the current state is directly influenced by the observed Q value of the next state–action combination. The Q value update [34] is represented in Eq. (13):

$$W(s_i, a_i) = W(s_i, a_i) + \alpha \cdot (d_t + \gamma \cdot W(s_{i+1}, a_{i+1}) - W(s_i, a_i)) \quad (13)$$

where γ indicates the discount factor which controls the future rewards to be used in the current update, α is the learning rate to control the step size, $W(s_i, a_i)$ represents the Q value in state s_i to take action a_i , immediate reward obtained is denoted as d_t and $W(s_{i+1}, a_{i+1})$ represents the Q value in next state s_{i+1} to take action a_{i+1} .

The reward d_t is defined in Eq. (14):

$$d_t = \beta_1 [f(s_i) - f(s_{i+1})] + \beta_2 v_t \quad (14)$$

where v_t is a term that promotes diversity and $f(\cdot)$ is the fitness function. ϵ -greedy policy has been used to balance exploration and exploitation as shown in Eq. (15)

$$v_t = \left\{ \text{randomaction}, \text{ifrand} < \epsilon; \arg \max_a W(s_i, a), \text{otherwise} \right\} \quad (15)$$

The decreasing ϵ across repetitions ensures the shift from exploration to exploitation. By learning optimal action strategies that efficiently direct the BEO's search dynamics, this integration promotes convergence.

3.4 Wrapper Classifier—Fuzzy KNN Using Bonferroni Mean

KNN (K-Nearest Neighbor) is a non-parametric supervised learning method that uses a majority vote of the nearest neighbors to determine a class sample. A membership degree is determined for every class in a fuzzy KNN algorithm [32], and the final classification is based on the highest membership degree. Let the training set with n number of samples as $C = \{C_1, C_2, C_3, \dots, C_n\}$. The test sample (x) membership degree in each class k is stated in Eq. (16) as follows:

$$\tau_k(x) = \frac{\sum_{i=1}^j q_{ki} \left(1/\|x - C_i\|^{\frac{2}{p-1}} \right)}{\sum_{i=1}^j \left(1/\|x - C_i\|^{\frac{2}{p-1}} \right)} \quad (16)$$

Here $p \in (1, \infty)$ and k th class i th observation membership is denoted as q_{ki} .

One of the operators for aggregation created by Bonferroni is the Bonferroni mean operator [33]. It permits repeated comparisons among the input arguments and can observe relationships between them. Let $\{C_1, C_2, C_3, \dots, C_n\}$, $C_i \in [0, 1]$ with parameters $m, p \geq 0$ and at least one $C_i \neq 0$. Equation (17) represents the Bonferroni mean of C_i .

$$B^{m,p}(C) = \left(\left(\frac{1}{n} \right) \sum_{i=1}^n C_i^m \left(\frac{1}{n-1} \sum_{i,j=1, j \neq i}^n C_j^p \right) \right)^{\frac{1}{m+p}} \quad (17)$$

The following Eq. (18) defines the fitness function that improves accuracy of the classifier by reducing the number of features chosen:

$$\text{Fitness} = \frac{1}{Z} \sum_{i=1}^Z A_i - \alpha \times \frac{|F|}{|T|} \quad (18)$$

Here, number of selected features indicated as $|F|$, $|T|$ represents the total features, A_i is the classification accuracy, $\alpha \in [0, 1]$ is the parameter to balance the chosen features and accuracy.

4 Proposed Model

The presence of an ovarian cysts and an excess of testosterone are the characteristics of PCOS. Despite the precise etiology is yet uncertain, it is induced by a combination of genetic and environmental factors. The common symptoms of PCOS include obesity, acne, type 2 diabetes, infertility and high risk of cardiovascular disease [10, 16]. In order to manage symptoms, prevent complications, and enhance the general well-being of women with PCOS, machine learning algorithms can be extremely beneficial by examining a variety of clinical, hormonal and lifestyle aspects to find pattern that can point to PCOS [2, 10, 11, 16–18]. In this article, the proposed work (Fig. 1.) consists of two main phases:

1. Hybrid filter-wrapper method using EF-RLBEO (Ensemble Filter—Reinforcement Learning-based enhanced Binary Equilibrium Optimizer) approach to identify the pertinent features
2. Machine Learning model to classify PCOS using optimal and stable features

4.1 Hybrid Feature Selection

Hybrid FS is a method that blends the filter and wrapper methods. In this article, the ensemble filter approach is initially developed to decrease the search space and dimensionality. Second, an improved (BEO) Binary Equilibrium Optimizer using Reinforcement Learning has been employed to achieve optimal reduced feature set. To avoid the local optima and increase convergence speed of BEO algorithm, Reinforcement Learning-based on-policy algorithm SARSA has been used.

4.1.1 Ensemble Filter Method and Its Stability Assessment

Ensemble filter method combines the result of individual filter method to improve the model accuracy. From a larger pool of available data, these techniques focus on extracting a smaller number of relevant features. In this article, individual methods, such as chi-square, ReliefF, Pearson, and Mutual Information, have been used. Since

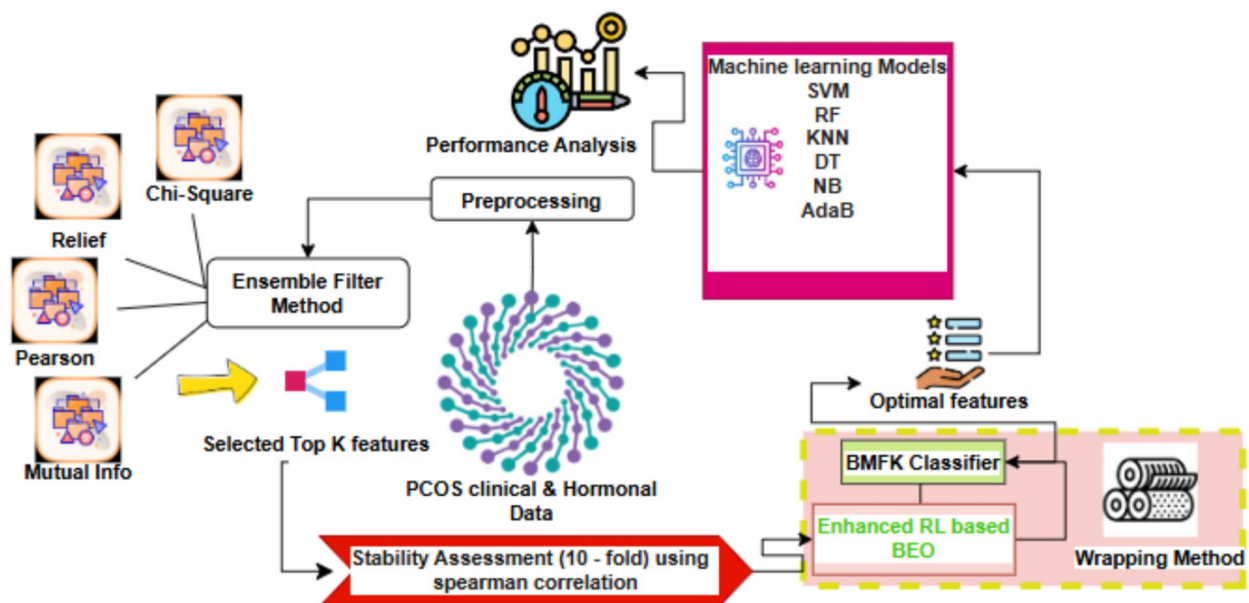


Fig. 1 Proposed classification model using Novel EF-RLBEO feature selection method

each of these methods assigns rank to each feature based on its own evaluation process, important features are less likely to be lost. An ensemble is a method which combines multiple FS methods to produce better results than an individual method. In order to assess the robustness of the features obtained by ensemble filter method, Stability is used as a performance metric. The feature selection methods must produce similar results even when there is a small modification in the data. To ensure the stability of each individual and ensemble filter method, the feature ranking in each iteration has been compared using tenfold cross-validation.

To assess the stability of individual filter and ensemble method with different threshold values, the Spearman rank correlation is employed. The stability score of individual filter method and ensemble method is given in Table 2. It is observed that the highest stability score has been attained by ensemble filter method than individual methods.

4.1.2 Wrapper Method–RLBEO (Reinforcement Learning (SARSA)-Based Enhanced Binary Equilibrium Optimizer)

Even BEO algorithm works well, it may get trapped in local optima and converge speed. Although previous research has investigated ways to improve the Equilibrium Optimiser (EO) using several metaheuristic hybrids and static parameter adjustment, EF-RLBEO provides several distinct innovations that together contribute to its improved performance. By ensuring that only high-quality, complementary characteristics enter the stochastic search, RL-driven EO synergy reduces noise and speeds up convergence. Also, SASRA agents learn to adjust the BEO's exploration coefficient based on the reward points earned from population diversity and fitness gains. This enables EF-RLBEO to keep its search trajectory balanced.

From this approach, the feature that received the highest reward points is considered for selecting the final optimal feature subset. For the subset evaluation, BMFK classifier has been employed due to its ability to handle the class imbalance problem efficiently [5]. Algorithm 1 describes the enhanced BEO using SARSA.

Algorithm 1 RLBEO (Reinforcement Learning (SARSA) based enhanced Binary Equilibrium Optimization).

```

Initialize the particles population with reduced feature set obtained from ensemble filter method
Initialize the objective function to evaluate the classification accuracy using BMFK
Determine the starting population's objective function
Initialize constants:  $G_p = 0.5$   $b1 = 2$  and  $b2 = 1$ 
for  $Irv = 1$  to  $Irv_{max}$ :
    for  $i = 1$  to  $n$ :
        Determine particles initial objective function
        Obtain  $C_{es1}, C_{es2}, C_{es3}, C_{es4}$  the top four candidates
    end for
    Calculate  $C_{es(avg)}$ , average of top four candidates using Eq. (6)
    Build  $C_{es,pool}$ , equilibrium pool using five candidates using Eq. (5)
    (if  $Irv > 1$ ), achieve memory saving
    Using Eq. (8), set time  $t$ 
    for  $i = 1$  to  $n$ :
        From  $C_{es,pool}$  equilibrium pool, randomly select any one candidate
        Calculate exponential  $X$  using Eq. (7)
        Calculate generation rate  $G$  using Eq. (9)
        Compute the term,  $C$  concentration using Eq. (12)
        Apply SARSA Eq. (13) to improve the search space and balance exploration
        and exploitation
        Using Eq. (18), determine the particles fitness value
        Based on the highest fitness value, find the best particles
    end for
end for

```

4.1.2.1 Influence of SARSA Updates on BEO Search In this article, SARSA is integrated into BEO approach for the efficient search process. Instead of choosing moves randomly, SARSA uses the rewards (fitness gains) to determine which actions (bit flips or binary updates) are more beneficial. This process guides the BEO search process to select the optimal features. The process involved in applying SARSA on BEO search is given in Algorithm 2.

In EF-RLBEO, the SARSA agent dynamically adjusts the exploration coefficient (ϵ) of the Binary Equilibrium Optimizer (BEO) based on incoming rewards, thereby steering the search trajectory. Let A, B, and C be three candidate subsets in a feature subset search space. BEO samples A first, calculates fitness $f(A)$, and then switches to B. This is tracked by SARSA as ($s_i = A$), $a_i = \text{transition_to_B}$. SARSA raises $W(A, a_i)$ in accordance with the reward $d_t > 0$ if $f(B) > f(A)$, which results in a greater exploration coefficient in the following phase. In contrast, SARSA might lower (ϵ) to encourage exploitation around the best solution at this point if $f(B) \leq f(A)$.

Algorithm 2 Influence of SARSA updates on BEO search process.

```

Initialize the SARSA Q-table (W) and population P
for i in 1:I do
  for every solution s in P, perform
    select action  $a_i$  (adjust  $\epsilon$ ) on  $W(s_i, \cdot)$  using  $\epsilon$ -greedy
    update BEO coefficient ( $\epsilon$ ) = a
    generate new candidate s' through BEO with ( $\epsilon$ )
    compute reward  $d_t = \beta_1 [f(s_i) - f(s_{i+1})] + \beta_2 v_t$ 
    select the next action  $a_{i+1}$  on  $W(s_{i+1}, \cdot)$  using  $\epsilon$ -greedy
     $W(s_i, a_i) = W(s_i, a_i) + \alpha \cdot (d_t + \gamma \cdot W(s_{i+1}, a_{i+1}) - W(s_i, a_i))$ 
     $s_i \leftarrow s_{i+1}$ 
  end for
end for

```

4.2 Machine Learning Classifiers

The optimal features obtained using hybrid filter-wrapper method are used as an input to classifier models. In this article, classifiers, such as SVM (Support Vector Machine), KNN (K-Nearest Neighbor), DT (Decision Tree), RF (Random Forest), NB (Naïve Bayes), and AdaB (AdaBoost), are implemented. Accuracy, Precision, Recall, F1 score, and AUC score are used to evaluate the individual models performance. The metrics are denoted as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \quad (19)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (20)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (21)$$

$$\text{F1 Score} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (22)$$

5 Experimental Study

This section briefly describes the evaluation of proposed method using PCOS dataset. The description and pre-processing of PCOS dataset is given in Sect. 5.1 and Sect. 5.2 describes the parameter setting & metrics used in each evaluation phase. The experimental results of the proposed model against other methods are given from Sect. 5.3, 5.4, 5.5, 5.6 and the ablation experiment conducted was presented in Sect. 5.7.

5.1 Dataset Description and Preprocessing

From Kaggle repository, publicly available PCOS dataset with clinical and physical characteristics is used [35]. The dataset consists of 541 patient records with 43 features. Initially, from the feature set, unwanted attributes like (SI.No) and (Patient File No) have been removed. Then, mean values are used to fill missing values of the relevant columns. Gaussian noise injection is applied to address the overfitting issue and make the model generalize better by making it learn to handle small variations in the input.

5.2 Parameter Setting and Metrics

In this article, three different phases of experiments have been conducted. As an initial phase, ensemble filter method is implemented to reduce the dimensionality and Spearman rank correlation is used to evaluate the robustness ensemble filter and individual methods. As the second phase, a wrapper method with RLBO is implemented. The results are compared with metaheuristic methods, such as PSO, GA, and GWO, using performance metrics includes mean fitness, number of features selected, accuracy and best fitness. In the third phase, classifiers, such as SVM, RF, DT, NB, KNN, and AdaBoost, have been implemented and evaluated using the metrics including accuracy, precision, recall, F1 score, and AUC score. Table 1 depicts the parameter setting used in each phase.

5.3 Ensemble Filter Methods with Stability Assessment

The Ensemble filter method combines the features obtained from individual filter method. In this work, top – k features identified by individual methods, such as Chi-square, Relief, Mutual Information, and Pearson, have been combined to find the optimal pertinent features. Additionally, the Spearman rank correlation is used to evaluate the robustness of features identified by ensemble and each individual method. Table 2 represents the stability score obtained by ensemble filter and individual methods with different optimal threshold values, such as 17, 23, 28, and 33, using tenfold cross-validation. The highest stability score of 92.20% has been obtained by Ensemble filter method with threshold of 17.

Table 1 Parameter settings

Methods	Values of parameter used
Common parameters used	N: Population Size: 10 (Max no. of iterations): GA: 10, PSO: 10, GWO: 5 5: k value of BMFK and KNN 2: BMFK—Fuzzy Strength Parameter in (m)
(PSO) Particle swarm optimization	Velocity Maximum (v_{max}): 0.1 Inertia Weight (w): 0.9 ($c1$ & $c2$): Constant value of Acceleration:0.5 each
(GA) Genetic algorithm	Crossover Rate: Implemented using cross point and offspring generation 0.1: Mutation Rate
Grey wolf optimization (GWO)	Convergence Parameter: Decreases from 2 to 0 over iterations non-linearly
Proposed (EF-RLBO)	BMFK Parameters: $n_{neighbors}$: 5, m : 2, p : 2, q : 2 BEO parameters: exploration coefficient $e = 0.9 \rightarrow 0.1$ (over 100 Iterations) RL: Learning Rate (α) = 0.1, Discount Factor (γ) = 0.9, Reward Weights: $f = 0.7$ (fitness gain) and $v_t = 0.3$ (diversity term)

Table 2 Stability Assessment of Individual Filter and Ensemble Filter Method using Spearman Rank Correlation

Methods	Threshold = 17	Threshold = 23	Threshold = 28	Threshold = 33
Relief	91.52	88.42	85.99	83.22
Chi—Square	91.00	89.36	87.05	82.96
Mutual info	90.25	88.89	88.03	82.57
Pearson	91.90	91.90	90.54	90.88
Proposed(Ensemble)	92.20	89.77	87.62	81.54

5.4 Results of the Proposed EF-RLBEO Approach with and Without Hybridization

The results obtained by the proposed EF-RLBEO approach with and without hybridization are presented in Table 3. The BMFK classifier is used as learning model. It can be seen that 96.31% improved accuracy has been achieved by the proposed hybrid EF-RLBEO with BMFK classifier than the accuracy of 82.61% and 82.33% produced by without hybridization i.e., RLBEO (wrapper) and EF(ensemble filter) alone. Further, the result of proposed hybrid method has been compared with EF + BEO using BMFK classifier and achieved lower accuracy of 81.51%, which shows the importance of hybridizing Ensemble Filter and enhanced wrapper method for the better model performance. Also, less accuracy of 81.13% has been achieved without EF and RLBEO i.e., all the features to BMFK classifier, which emphasizes the need for enhanced feature selection process to obtain better accuracy.

Additionally, statistical analyses have been performed on the cross-validated accuracy scores obtained over 10 repeated runs for each feature selection approach (EF alone, BEO alone, and EF-RLBEO) to validate the performance enhancements of EF-RLBEO over baseline methods. The outcomes of the Friedman test with Nemenyi post hoc, Wilcoxon signed-rank tests, and one-way repeated measures ANOVA with Tukey HSD post hoc are shown below. The results of pairwise comparisons of accuracy scores between each baseline technique and EF-RLBEO using the Wilcoxon signed-rank test are displayed in Table 4. EF-RLBEO significantly outperforms the baseline approaches as indicated by significant p values (< 0.05).

Table 5 presents the Friedman test findings for the three techniques. The results showed substantial differences, with $\chi^2(2) = 12.53$ and $p = 0.002$. The average rank of EF-RLBEO is significantly higher according to post hoc Nemenyi analysis (critical difference = 0.75).

A one-way repeated measures ANOVA was conducted to compare mean accuracy scores. Table 6 shows the ANOVA summary, where SS defines Sum of Squares, DF denotes Degrees of Freedom, MS is the Mean Square, F denotes (F-statistic), and p is the probability value. This test was performed between and within groups. The results indicated a significant main effect between methods with F value of 10.45 and $p = 0.001$.

Table 3 Accuracy comparison of proposed EF-RLBEO with and without hybridization

Methods	Accuracy	Friedman mean rank	Rank
without EF filter & wrapper to BMFK (all features)	81.13	3.60	4
With EF and without RLBEO	82.33	3.20	3
With EF & with BEO	81.51	4.20	5
Without EF & with RLBEO + BMFK classifier	82.61	2.80	2
With EF & RLBEO + BMFK classifier (proposed)	96.31	1.20	1

Table 4 Wilcoxon Signed-Rank Test

Methods	Wilcoxon p value	Significant ($\alpha = 0.05$)
EF & EF-RLBEO	0.008	Yes
BEO & EF-RLBEO	0.015	Yes

Table 5 Friedman Test with Nemenyi Post hoc

Methods	Average rank	Critical difference	Significant
EF	2.5	0.75	No
BEO	2.2	0.75	No
EF-RLBEO	1.3	0.75	Yes

Table 6 One-Way Repeated Measures ANOVA

Source	SS	DF	MS	F	<i>p</i> value
Between groups	12.30	2	6.15	10.45	0.001
Within groups	10.60	18	0.59		

Table 7 Tukey HSD Post hoc results

Methods	Mean difference	Adjusted <i>p</i> value	Significant
EF & BEO	−1.3	0.045	Yes
EF & EF-RLBEO	−2.8	0.002	Yes
BEO & EF-RLBEO	−1.5	0.015	Yes

Table 8 Comparison of proposed approach with other metaheuristic algorithms

Methods	Best fitness	Mean fitness	Accuracy	Friedman mean rank
GA- BMFK	85.18	83.25	87.03	2.33
PSO- BMFK	84.25	83.36	86.11	3.33
GWO- BMFK	83.33	82.26	84.25	3.5
EF-RLBEO- BMFK(Proposed)	92.59	83.53	96.31	1.00

Table 7 presents post hoc Tukey HSD comparisons. According to the Tukey HSD post hoc test, EF-RLBEO is statistically superior to both EF and BEO.

5.5 Performance Analysis of the Proposed EF-RLBEO Approach with other Feature Selection Method Using Metaheuristic Approach

The comparison of proposed EF-RLBEO with other FS method using metaheuristic approach is depicted in Table 8 and Fig. 2. The metaheuristic methods including Swarm optimizer [10], Grey Wolf Optimizer (GWO) [11], and Genetic Algorithm (GA) [21] have been implemented using BMFK as classifier. Mean fitness, best fitness, number of features selected, and accuracy have been used as performance metrics. The proposed EF-RLBEO produced the highest best fitness value of 92.59% and GA-BMFK stood second, achieved 85.18% best fitness value. In terms of Mean fitness, proposed EF-RLBEO method produced 83.53% as the highest followed by PSO-BMFK with 83.36% as the second highest value. Regarding accuracy, it is observed that the proposed EF-RLBEO outperformed well and attained the highest accuracy 96.31% and GA – BMFK has obtained 87.03% as the second highest accuracy. The Friedman mean rank is applied to examine the rank of each method. The proposed EF-RLBEO attained the top rank of 1.00 followed by GA-BMFK stood second rank of 2.33.

The number of features selected using proposed EF-RLBEO and PSO, GWO, and GA is given in Fig. 3. The highest classification accuracy of 95.62% has been attained using 13 optimal features produced by proposed EF-RLBEO (Table 11), which is the highest than other methods. These selected features belong to various categories, such as Demographic: Age (yrs), Anthropometric: (BMI (kg/m²), Waist:Hip Ratio as central obesity indicator, Hormonal: (AMH (ng/mL), PRG (ng/mL), FSH/LH Ratio), Ovarian Morphology: Follicle

Fig. 2 Best fitness, Mean fitness and accuracy comparison of proposed with other methods

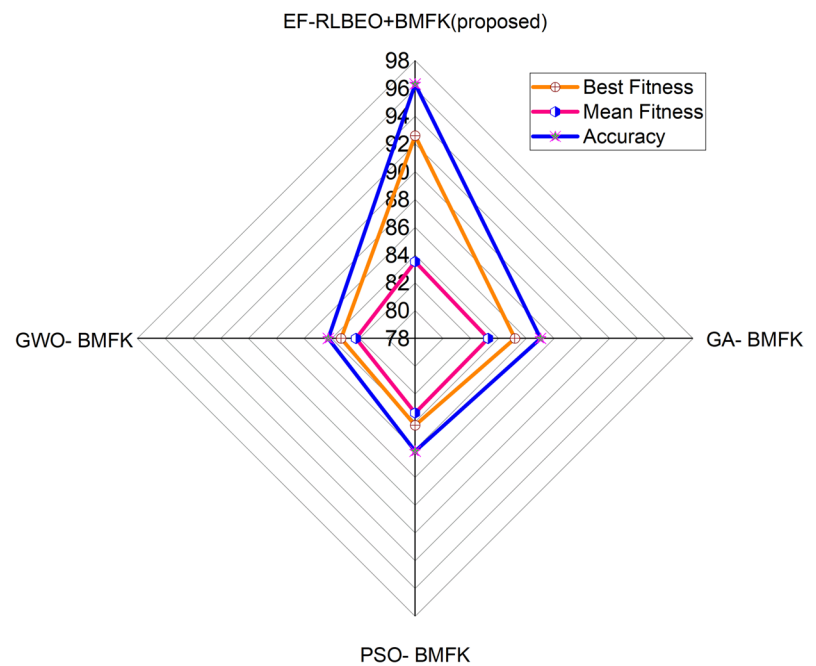
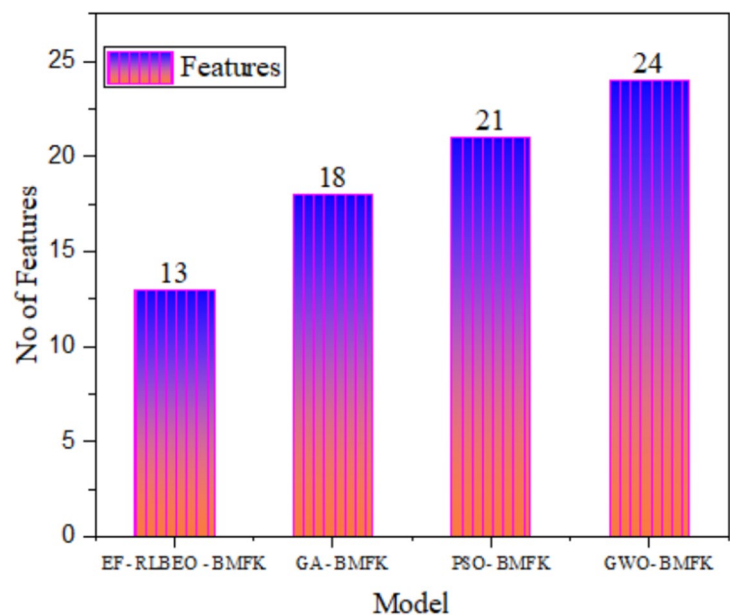


Fig. 3 Features selected by proposed and other methods



No. (L), Follicle No. (R), Avg. F Size (L) (mm), Avg. F Size (R) (mm), Reproductive Health: Endometrium (mm), Clinical Sign: Skin Darkening (Y/N), and Lifestyle: Fast Food (Y/N). Additionally, to quantify each feature's contribution to final decision, SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations) are used in the EF-RLBEO-based PCOS classifier to ensure transparency and reliability.

Figures 4 and 5 present the results of SHAP and LIME. In SHAP, Features are ranked according to their overall influence on model output, which is determined by their mean absolute SHAP value. It is noted from Fig. 4 that Hormonal feature AMH obtained the highest SHAP value of 0.21 followed by FSH/LH ratio and BMI. With an ovarian morphological feature including Follicle No (L) with 0.19, Follicle No (R) with 0.16 obtained the highest SHAP values and contributed more for model improved performance.

Fig. 4 Global SHAP values of selected features

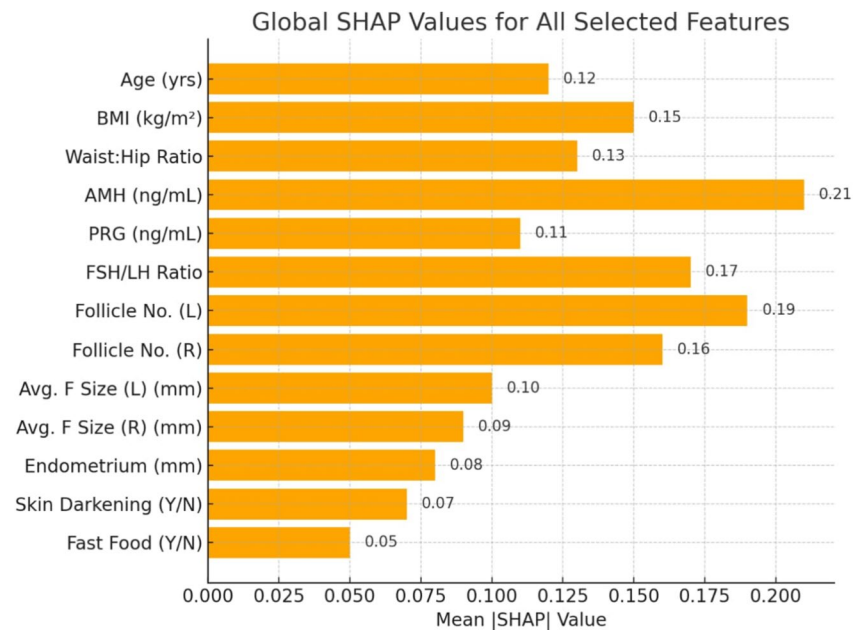
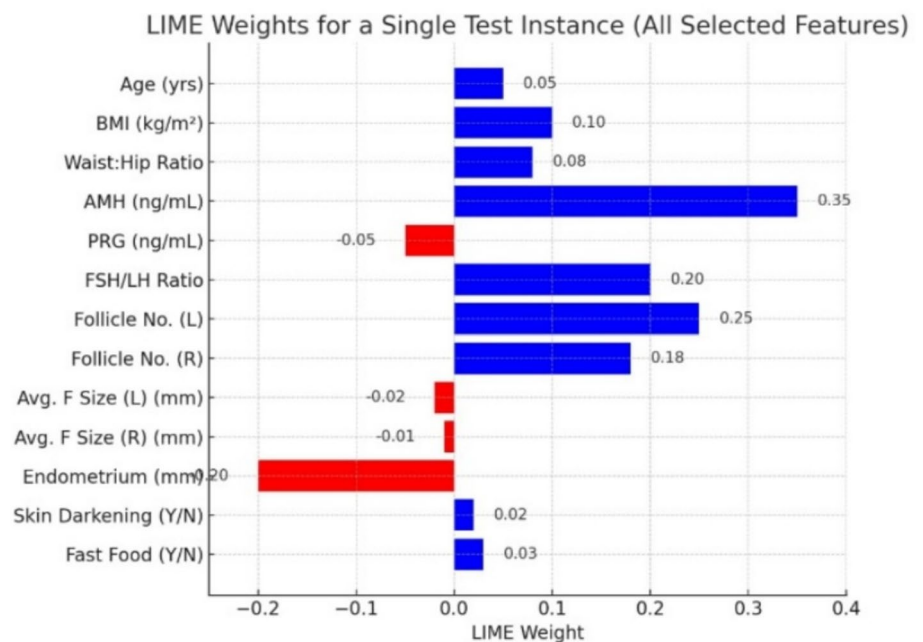


Fig. 5 LIME weights of selected features



Further, LIME explanations have been generated for local interpretability, and these are shown in Fig. 5. It is observed that AMH has a consistent influence on the model's prediction, peaking at 0.35 (AMH), 0.25 (Follicle No (L)), 0.20 (FSH/LH ratio), 0.18 (Follicle No (R)), and BMI of 0.10. This notable impact implies that the model heavily relies on this specific attribute when making decisions.

Figures 6 and 7 display the convergence trends of proposed EF-RLBEO and other metaheuristic methods PSO, GWO, and GA based on mean fitness and best fitness values. It is observed from the results that the proposed EF-RLBEO is exceptionally superior to its convergence behavior as it settles with the highest fitness value in early stages and produces the rapid convergence which remains constant for 100 iterations. It shows that the proposed EF-RLBEO method is more stable and can avoid local optima, the most often problem of metaheuristic approach. Also, the constant fitness value suggests that the proposed method handles exploration

Fig. 6 Convergence trends of proposed & other metaheuristic methods based on Best fitness values

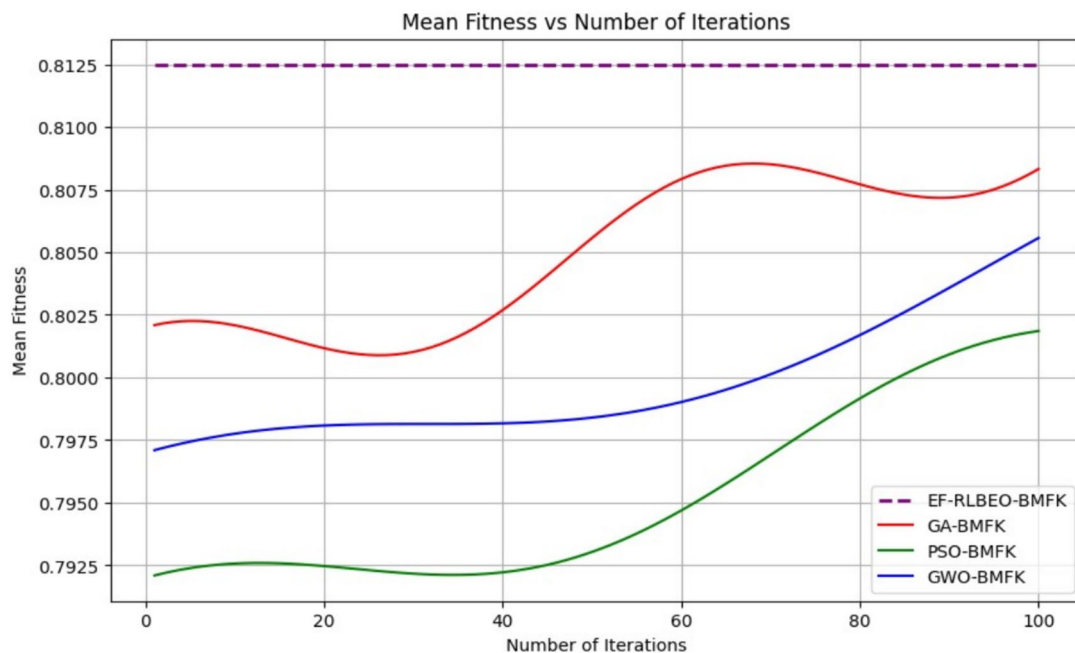
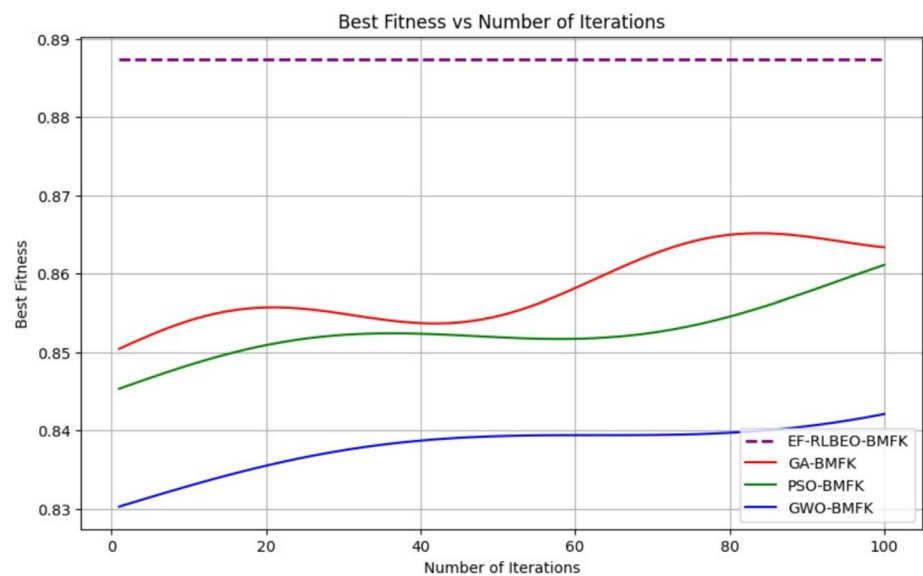


Fig. 7 Convergence trends of proposed & other metaheuristic methods based on Mean fitness values

and exploitation efficiently. Additionally, less iteration needed for convergence point leads to increased computational efficiency. In contrast, GA, PSO, and GWO approaches fluctuate more and number of iteration to attain optimal solution is also high.

To evaluate the computational efficiency of EF-RLBEO, average runtime, number of iterations to convergence, and peak memory usage for each feature selection method over 10 runs on the Kaggle PCOS dataset (with 541 samples and 43 features) were measured. Table 9 summarizes the results.

Due to the RL agent computations, EF-RLBEO has a substantial overhead (~30–45% longer runtime) in comparison to normal BEO hybrids. However, it converges in fewer iterations and achieves greater performance, indicating a favorable trade-off between accuracy and computational cost.

Table 9 Computational measures of proposed and other methods

Method	Avg. runtime (s)	Iterations to converge	Peak memory (MB)	Description
EF only	8.5	N/A	85	Filter stage only
BEO only	120.3	60	210	Standard BEO
GWO-BEO	140.7	75	215	Grey Wolf hybrid
PSO-BEO	130.5	70	212	PSO hybrid
EF-RLBEO	180.2	65	225	Ensemble + RLBEO

5.6 Individual Classifier Performance Analysis -without and with Features Selected by EF-RLBEO Feature Selection Method

Figure 8a and b displays the accuracy comparison of machine learning classifiers without and with proposed feature selection method. It can be observed from Fig. 8a, without using features selected by proposed EF-RLBEO, 89.12% of accuracy has been achieved by Random Forest, followed by AdaBoost with 87.88%. Conversely, from Fig. 8b, it shows that the highest accuracy of 95.62% has been achieved by Random Forest using 13 most pertinent features selected by proposed EF-RLBEO and AdaBoost stood second with 92.38% of accuracy, which shows the importance of selecting reduced pertinent feature set for the better model performance.

Tables 10 and 11 depict the individual classifier analysis without and with proposed feature selection method. In this article, high performed classifier, such as SVM (Support Vector Machine), KNN, DT (Decision Tree), RF (Random Forest), Naïve Bayes, and AdaBoost, has been used for PCOS classification. Table 10 depicts the classification performance of individual classifier model without using proposed EF-RLBEO results. It is seen

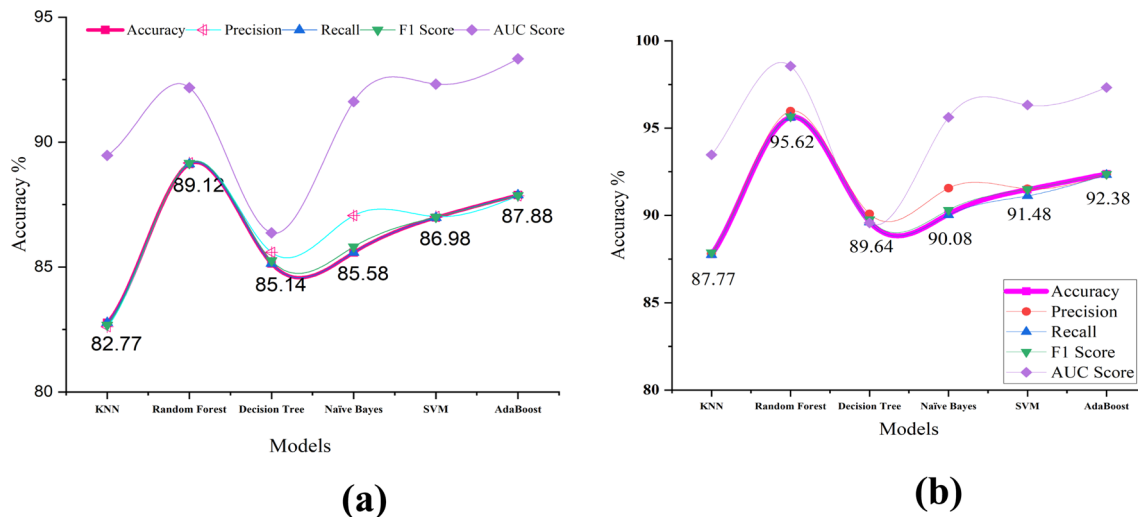


Fig. 8 Accuracy Comparison of Individual Models **a** without using selected features **b** with using selected features using proposed EF-RLBEO

Table 10 Performance analysis of classifiers without using optimal features produced by proposed EF-RLBEO

Methods	Accuracy	Precision	Recall	F1 score	AUC score
AdaBoost	87.88	87.85	87.88	87.87	93.33
KNN	82.77	82.61	82.77	82.68	89.47
Random forest	89.12	89.17	89.12	89.16	92.18
Decision tree	85.14	85.59	85.14	85.26	86.37
Naïve Bayes	85.58	87.06	85.58	85.81	91.62
SVM	86.98	87.02	86.98	87.00	92.32

Table 11 Classification performance analysis of classifiers using optimal features produced by proposed EF-RLBEO

Methods	Accuracy	Precision	Recall	F1 score	AUC score
AdaBoost	92.38	92.35	92.32	92.37	97.33
KNN	87.77	87.81	87.74	87.88	93.47
Random forest	95.62	95.97	95.61	95.66	98.55
Decision tree	89.64	90.09	89.61	89.76	89.57
Naïve Bayes	90.08	91.56	90.02	90.31	95.62
SVM	91.48	91.52	91.12	91.50	96.32

that the highest AUC score of 92.18%, precision of 89.17%, Recall of 89.12% and 89.16% of F1 score has been achieved by Random Forest. AdaBoost stood second, achieved 87.85% precision, 87.88% Recall, 87.87% F1 score and AUC of 93.33%. Figure 9 shows the RoC (Receiver Operating Characteristic) plot of individual model without using FS. It is noticed that true positive rate attained by RF (Random Forest) is higher than other classifiers. Table 11 displays the individual classifier performance using features produced by EF-RLBEO. It is observed that improved AUC score of 98.55% was obtained by Random forest using 13 features, and less AUC of 89.57% was obtained by Decision Tree. Figure 10 shows the RoC plot of individual model with FS. It is observed that true positive rate attained by Random Forest using optimal feature subset produced by proposed EF-RLBEO surpassed the other classifiers.

Further, Women's Health Boston PCOS dataset from the UCI Machine Learning Repository was used to test the framework's generalizability and robustness. There are 36 clinical characteristics and 300 samples in this dataset. The performance was compared to baseline approaches (EF only, BEO only) using the same EF-RLBEO procedure. Table 12 provides a summary of the BMFK classifier's performance parameters, including accuracy, precision, recall, F1 score, and AUC.

The findings confirm the generalizability and robustness of EF-RLBEO across various clinical data distributions by showing that it continuously outperforms the individual EF and BEO methods on an independent PCOS dataset with the highest accuracy of 95.8%.

Table 13 depicts the comparison of the proposed model performance with existing work. Gosain et al. [17] have obtained 90.24% accuracy using six classifiers AdaB, NB, LR, DT, RF, and SVM with 30 features selected by

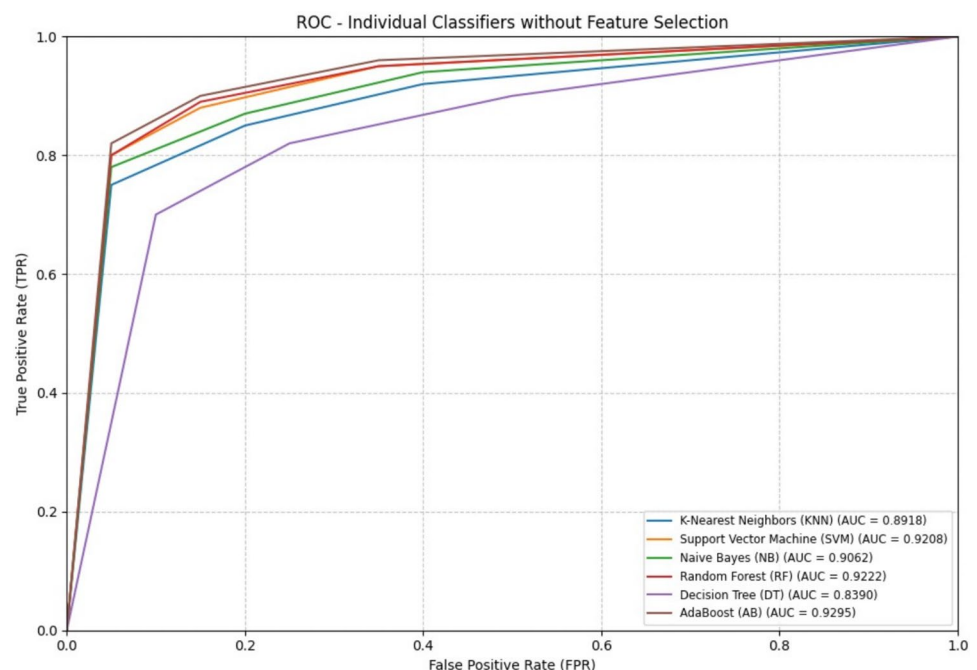
Fig. 9 ROC plot of classifier models without FS

Fig. 10 ROC plot of classifier models with FS

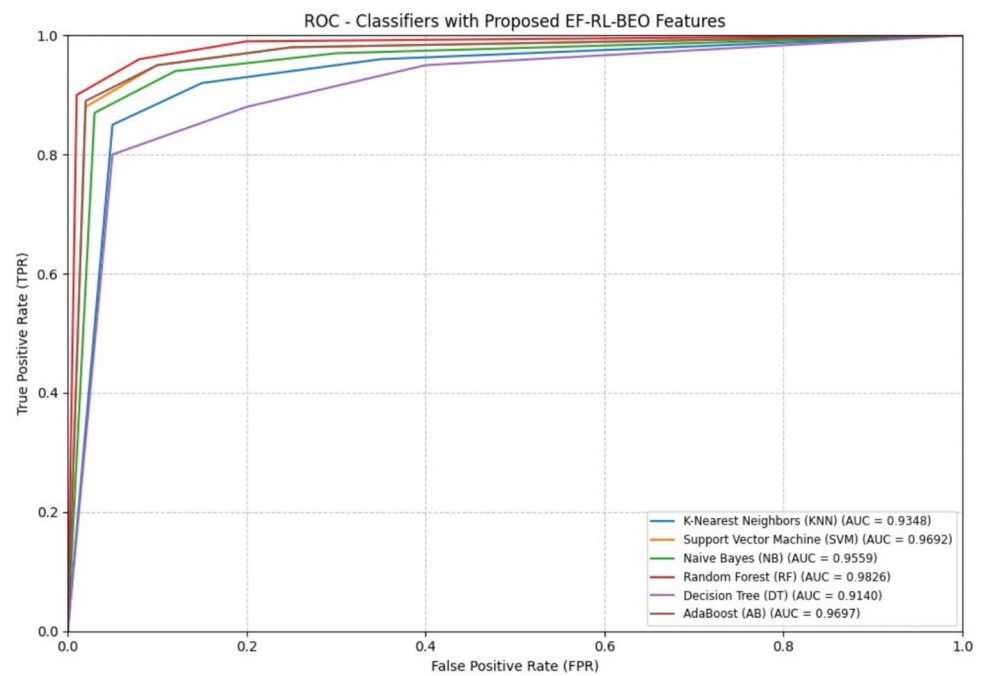


Table 12 Performance analysis of proposed EF-RLBEO vs other baseline methods using Women's Health Boston PCOS dataset

Methods	Accuracy	Precision	Recall	F1 score	AUC score
EF only	92.1	91.8	92.5	92.1	0.94
BEO only	93.4	93.0	93.6	93.3	0.95
EF-RLBEO	95.8	95.5	96.0	95.7	0.97

Table 13 Accuracy comparison of EF-RLBEO (proposed model) with other existing results

Source	Classifier model	FS(Feature selection) method	No.of features	Accuracy %
Anjana Gosain et.al., [17]	AdaB, NB, LR, DT, RF,SVM	Backward Feature Elimination (BFE)	30	90.24
Tiwari et al., [18]	SVM, LR, RF, AdaB, DT, KNN, XGBoost, CatBoost	Pearson Correlation	15	93.25
Proposed model	KNN, LR, SVM, DT, RF, NB, AdaBoost	Proposed (EF-RLBEO)	13	95.62

BFE (Backward Feature Elimination). Tiwari et al. [18] attained 93.25% accuracy with eight different classifiers using 15 features identified by Pearson correlation. The proposed model has obtained accuracy of 95.62% with 13 features selected by the proposed EF-RLBEO feature selection method outperformed well than existing works.

5.7 Ablation Experiments

This section details three distinct ablation experiments that were carried out to assess the model's performance based on the number of features selected by various optimization techniques. This article utilized different optimization techniques to identify most pertinent features, such as GA-BMFK, PSO-BMFK, GWO-BMFK and

EF-RLBEO (proposed). The experimented results of the classifier models against distinct features obtained by these four optimization algorithms are given in Table 14.

- With 24 features produced by GWO – BMFK, Random Forest has achieved the accuracy of 87.25%, which is comparatively 8.37% lower than model trained with proposed EF-RLBEO.
- With 21 features produced by PSO – BMFK, Random Forest has attained accuracy of 90.79%, which is comparatively 4.83% lower than model trained with proposed EF-RLBEO.
- Whereas, with reduced feature set of 18 obtained by GA – BMFK, Random Forest obtained improved accuracy of 93.28%, which is 2.34% lower than proposed EF-RLBEO results.
- From the experiments results, it is observed that the model performance was less when trained with more number of features. This may occur due to the irrelevant features and overfitting issue.

6 Conclusion

This article proposes EF-RLBEO, a hybrid filter-wrapper method for the accurate and prompt diagnosis of PCOS. Initially, Ensemble Filter (EF) method is used to obtain the reduced feature set based on ranking and the Spearman correlation is employed to assess the stability of the FS methods. Further, the reduced feature set obtained using EF method is given as an input to RLBEO (Reinforcement Learning-based Binary Equilibrium Optimizer) for finding the smaller optimal feature subset. Here, Reinforcement Learning used on-policy algorithm SARSA to improve the ability of exploration capability and address the local optima issue. BMFK is used as evaluation model in wrapper method and achieved the highest accuracy of 96.31% than other metaheuristic methods. Then, classifier models, such as Random Forest (RF), Decision Tree (DT), Naïve Bayes (NB), Support Vector Machine (SVM), KNN (K-Nearest Neighbor), and AdaB (AdaBoost), have been implemented using optimal features obtained by EF-RLBEO. The performances of the classifier models are evaluated using metrics, such as accuracy, precision,

Table 14 Ablation Experimentation Results of Classifiers with number of features produced by other metaheuristic approaches

Optimization algorithm (Number of Features)	Classification models	Accuracy	Precision	Recall	F1 Score	AUC Score
GA – BMFK (18)	AdaBoost	91.76	92.46	92.11	92.62	93.15
	KNN	86.89	86.45	86.73	86.77	90.37
	Random forest	93.28	93.17	93.06	93.27	95.42
	Decision tree	92.86	90.32	89.67	90.14	90.95
	Naïve Bayes	89.18	89.05	89.03	89.63	91.65
	SVM	91.65	91.32	91.24	91.89	92.16
PSO – BMFK (21)	AdaBoost	88.91	89.24	89.11	89.14	90.17
	KNN	83.78	83.45	83.57	83.88	85.90
	Random forest	90.79	91.23	91.56	91.38	92.65
	Decision tree	87.60	87.14	87.59	87.67	88.12
	Naïve Bayes	87.92	86.96	86.45	87.12	90.56
	SVM	88.45	89.67	88.47	88.22	91.27
GWO – BMFK (24)	AdaBoost	86.59	86.42	86.28	86.39	87.31
	KNN	81.83	81.17	81.44	81.58	87.70
	Random forest	87.25	87.30	87.14	88.65	89.73
	Decision tree	85.92	85.71	85.64	85.77	86.98
	Naïve Bayes	86.14	85.44	85.67	86.26	88.66
	SVM	84.67	84.32	84.39	84.17	86.91

AUC score, F1 score, and recall. The highest accuracy of 95.62% has been achieved by Random Forest (RF), surpassed other models. Through this research, a pertinent feature that serves as the primary marker for PCOS diagnosis has been identified. This would help medical professionals to diagnose the diseases more rapidly and accurately. Despite EF-RLBEO encouraging performance, it has some limitations. SARSA integration raises the computational cost and necessitates meticulous tuning of several hyper-parameters. Also, only numerical clinical dataset was used to evaluate the approach, which limited its applicability to other data modalities. Furthermore, to verify robustness and repeatability, this work can be expanded in future with further extensive validation across various medical datasets. Also, it can be expanded to implement a method that automatically determines the threshold value for choosing the informative features from the group of filtering techniques.

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Data Availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

Ethical Approval This article contains no studies with human participants performed by authors.

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References

1. Oğuz, S.H., Boyraz, B.S., Ertürk, B., Yıldız, B.O.: The prevalence, phenotype and cardiometabolic risk of polycystic ovary syndrome in treatment-naïve transgender people assigned female at birth. *Endocrine* **84**(1), 287–292 (2024)
2. Bharati S., Podder P., Mondal M.R. H. (2020). Diagnosis of Polycystic Ovary Syndrome Using Machine Learning Algorithms. *IEEE Region 10 Symposium (TENSYP)*, Dhaka, Bangladesh, 1486–1489.
3. Miles, K. S. (2023). Our Pearls Matter: PCOS through the Lens of Women of Color and White Women.
4. Bhat, S. A. (2021). Detection of polycystic ovary syndrome using machine learning algorithms. PhD diss., Dublin, National College of Ireland.
5. Vommi, A.M., Battula, T.K.: A hybrid filter-wrapper feature selection using Fuzzy KNN based on Bonferroni mean for medical datasets classification: A COVID-19 case study. *Expert Syst. Appl.* **218**, 119612 (2023)
6. El-Rashidy, N., Tarek, Z., Elshewey, A.M., Shams, M.Y.: Multitask multilayer-prediction model for predicting mechanical ventilation and the associated mortality rate. *Neural Comput. Appl.* **37**, 1321–1343 (2025)
7. Elshewey, A.M., Alhussan, A.A., Khafaga, D.S., Elkenawy, E.-S., Tarek, Z.: EEG-based optimization of eye state classification using modified-BER metaheuristic algorithm. *Sci. Rep.* **14**, 24489 (2024)
8. Wolpert, D.H., Macready, W.G.: No free lunch theorems for optimization. *IEEE Trans. Evol. Comput.* **1**(1), 67–82 (1997). <https://doi.org/10.1109/4235.585893>
9. Abdel-Basset, M., Chang, V., Mohamed, R.: A novel equilibrium optimization algorithm for multi-thresholding image segmentation problems. *Neural Comput. Appl.* **33**, 10685–10718 (2021)

10. Mahesswari G. U., Maheswari P. U. (2024). SmartScanPCOS: A Feature-Driven Approach to Cutting-Edge Prediction of Polycystic Ovary Syndrome using Machine Learning and Explainable Artificial Intelligence. *Heliyon*.
11. Moral, P., Mustafi, D., Sahana S.K.: PODBoost: an explainable AI model for polycystic ovarian syndrome detection using grey wolf-based feature selection approach. *Neural. Comput. Appl.* **36**, 1–18 (2024)
12. Elkenawy, E.-S., Alhussan, A.A., Khafaga, D.S., Tarek, Z., Elshewey, A.M.: Greylag goose optimization and multilayer perceptron for enhancing lung cancer classification. *Sci. Rep.* **14**(1), 23784 (2024)
13. Tarek, Z., Alhussan, A.A., Khafaga, D.S., El-Kenawy, E.-S., Elshewey, A.M.: A snake optimization algorithm-based feature selection framework for rapid detection of cardiovascular disease in its early stages. *Biomed. Signal Process. Control* **102**, 107417 (2025)
14. Elshewey, A.M., Osman, A.M.: Orthopedic disease classification based on breadth-first search algorithm. *Sci. Rep.* **14**(1), 23368 (2024)
15. Ahmad, R., Maghrabi, A., Khaja, I.A., Maghrabi, A., Ahmad, M.: SMOTE-based automated PCOS prediction using lightweight deep learning models. *Diagnostics* **14**(19), 2225 (2024)
16. Aggarwal, S., Pandey, K.: Early identification of PCOS with commonly known diseases: Obesity, diabetes, high blood pressure and heart disease using machine learning techniques. *Expert Syst. Appl.* **217**, 119532 (2023)
17. Kumari, R., Singh, J., Gosain, A.: SmS: SMOTE-stacked hybrid model for diagnosis of polycystic ovary syndrome using feature selection method. *Expert Syst. Appl.* **225**, 120102 (2023)
18. Tiwari, S., Kane, L., Koundal, D., Jain, A., Alhudhaif, A., Polat, K., Zaguia, A., Alenezi, F., Althubiti, S.A.: SPOSDS: A smart polycystic ovary syndrome diagnostic system using machine learning. *Expert Syst. Appl.* **203**, 117592 (2022)
19. Mehr, D., Homa, H.P.: Diagnosis of polycystic ovary syndrome through different machine learning and feature selection techniques. *Heal. Technol.* **12**(1), 137–150 (2022)
20. Inan, M. S. K., Ulfath R. E., Alam F. I., Bappee F. K., Hasan R. (2021). Improved sampling and feature selection to support extreme gradient boosting for PCOS diagnosis. *IEEE 11th annual computing and communication workshop and conference (CCWC)*, 1046–1050.
21. Munjal, A., Khandia, R., Gautam, B.: A machine learning approach for selection of Polycystic Ovarian Syndrome (PCOS) attributes and comparing different classifier performance with the help of WEKA and PyCaret. *Int. J. Sci. Res* **2277**, 59–63 (2020)
22. Elmanakhly, D.A., Saleh M.M., Rashed E.: An improved equilibrium optimizer algorithm for features selection: methods and analysis. *IEEE Access* **9**, 120309–120327 (2021)
23. Too, J., Mirjalili, S.: General learning equilibrium optimizer: a new feature selection method for biological data classification. *Appl. Artif. Intell.* **35**(3), 247–263 (2021)
24. Awadallah, M.A., Hammouri, A.I., Al-Betar, M.A., Braik, M.S., Abd, M., Elaziz: Binary Horse herd optimization algorithm with crossover operators for feature selection. *Comput. Biol. Med.* **141**, 105152 (2022)
25. Vommi, A.M., Battula, T.K.: A binary Bi-phase mutation-based hybrid Equilibrium Optimizer for feature selection in medical datasets classification. *Comput. Electr. Eng.* **105**, 108553 (2023)
26. Varzaneh, Z.A., Hossein, S., Mood, S.E., Javidi, M.M.: A new hybrid feature selection based on improved equilibrium optimization. *Chemom. Intell. Lab. Syst.* **228**, 104618 (2022)
27. Sayed, G.I., Khoriba, G., Haggag, M.H.: A novel chaotic equilibrium optimizer algorithm with S-shaped and V-shaped transfer functions for feature selection. *J. Ambient. Intell. Humaniz. Comput.* **13**(6), 3137–3162 (2022)
28. Got, A., Moussaoui, A., Zouache, D.: Hybrid filter-wrapper feature selection using whale optimization algorithm: A multi-objective approach. *Expert Syst. Appl.* **183**, 115312 (2021)
29. Hammami, M., Bechikh, S., Hung, C.-C., Said, L.B.: A multi-objective hybrid filter-wrapper evolutionary approach for feature selection. *Memetic Comput.* **11**, 193–208 (2019)
30. Ghosh, M., Adhikary, S., Ghosh, K.K., Sardar, A., Begum, S., Sarkar, R.: Genetic algorithm based cancerous gene identification from microarray data using ensemble of filter methods. *Med. Biol. Eng. Compu.* **57**, 159–176 (2019)
31. Faramarzi, A., Heidarinejad, M., Stephens, B., Mirjalili, S.: Equilibrium optimizer: A novel optimization algorithm. *Knowl.-Based Syst.* **191**, 105190 (2020)
32. Keller, J.M., Gray, R.M., Givens, J.A.: A fuzzy k-nearest neighbor algorithm. *IEEE Trans. Syst. Man Cybern.* **4**, 580–585 (1985)
33. Bonferroni, C.: Sulle medie multiple di potenze. *Bollettino dell'Unione Matematica Italiana.* **5**(3–4), 267–270 (1950)
34. Sutton, Richard. (2018). *Reinforcement learning: An introduction*, A Bradford Book,
35. Dataset: <https://www.kaggle.com/datasets/prasoonkottarathil/polycystic-ovary-syndrome-pcos/data>. Accessed April 2025

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